



Featured Article

Longitudinal associations between social relationships and cognitive function among Chinese older adults: The mediating role of healthy lifestyles

Weibo Ma, PhD^a, Bei Wu, PhD^b, Ying Yu, PhD^c, Renyao Zhong, PhD^{a,*}

^a School of Public Administration, Faculty of Economics and Management, East China Normal University, Shanghai, China

^b NYU Aging Incubator and Hartford Institute for Geriatric Nursing, New York, USA

^c Shanghai University of Medicine and Health Sciences, Shanghai, China



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ABSTRACT

Background: We aimed to explore whether social relationships and cognitive function were reciprocally related and whether healthy lifestyles mediate their association.

Methods: This study included 3372 Chinese adults from the years 2014–2018 wave of the Chinese Longitudinal Healthy Longevity Survey.

Results: Baseline social relationships were negatively associated with a change in cognitive function ($\beta = -0.030$, $p = 0.043$). Participants with a 1 standard deviation (SD) increase in social relationships had an approximately 6% reduced risk of developing cognitive impairment. However, baseline cognitive function did not independently predict social relationships. The association between Δ social relationships and Δ cognitive function was partially mediated by Δ a healthy lifestyle ($B = 0.025$, 95% CI = 0.013–0.041).

Conclusions: The association between social relationships and cognitive function may be unidirectional. Policies that promote cognitive function based on social relationships would benefit by taking into account lifestyle factors.

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Introduction

Late-life cognitive impairment and dementia are considered major public health concerns due to their high prevalence rates and economic and social burden.¹ The total number of people with dementia is projected to reach 78 million by 2030 and 139 million by 2050 globally.² The number of patients with dementia in China accounts for approximately a quarter of the entire population with dementia worldwide.³ A prevalence of 6.0 and 15.5% for dementia and mild cognitive impairment, respectively, are found in people aged 60 years and older.⁴ The increasing global threat posed by dementia and the lack of established curative treatments makes it imperative to focus on risk reduction.

Social relationships refer to the connections between individuals and their social environment.⁵ Positive social relationships are the modifiable and convenient factors that can prevent cognitive decline. Kelly *et al.* suggested a framework to classify social relationships into three areas, namely, social activities or participation, social networks,

and social support.⁶ From the theoretical point of view, there are clear reasons to expect a correlation between social relationships and cognitive outcomes. According to cognitive reserve theory, participating in social and physical activities will stimulate the brain and help establish additional neuron resources or cognitive strategies to buffer the decline of cognitive ability.⁷ The social control hypothesis suggests that interacting with others in one's own social network may lead to positive and healthy behaviors.⁸ The other explanation is social support, in particular, may influence cognitive outcomes through its buffering effect on stress.⁹ Many studies have reported the relationship between social relationships or dimensions of social relationships and cognitive function.^{10–14} The meta-analysis of longitudinal cohort studies reported that poor social relationships are associated with cognitive decline.⁵ One prospective study found that a high level of social relationships reduces the risk of cognitive decline, especially active engagement in social activities and connected social networks.¹⁴ However, Kuiper *et al.* found that the association between social relationships and cognitive decline was not present or of a smaller magnitude among studies that adjusted potential confounders (i.e., depression, alcohol use, physical functioning).⁵ Thus, the research on the influence of social relationships on cognitive decline needs to be explored yet.

*Corresponding author at: School of Public Administration, Faculty of Economics and Management, East China Normal University, 3663 Zhongshan Road, Putuo District, Shanghai, 200062, China.

E-mail address: ryzhong297@hotmail.com (R. Zhong).

The Interactive Biopsychosocial Model emphasizes bidirectional causality and feedback, suggesting that social resources shape health, but health also shapes individuals' experiences of the social world.¹⁵ For example, a good partnership may promote cognitive health through increasing social interaction, while chronic marital strain may damage cognitive functioning.^{16,17} Corresponding to this, cognitive impairment also predicts lower participation in social and leisure activities.¹⁸ It is worth noting that if elderly individuals who have experienced cognitive decline begin to gain more social support, there may not be a straightforward decline in social relationships. This is, of course, based on the background of Western culture. In China, families are influenced by Confucianism, a traditional Chinese culture, and take it for granted to get the support of their families, as Confucianism emphasizes having an internal moral compass or conscience.¹⁹ The degree of social support may not change much with the decline of cognitive function. This inspires us to explore whether social relationships and cognitive impairment are mutually causal and consequential. To the best of our knowledge, there are no longitudinal research reports on the bidirectional association between social relationships and cognitive impairment in older people in China.

Social relationships express the full spectrum of lifestyles, including behaviors and norms that can lead to healthy or unhealthy outcomes.²⁰ The British longitudinal study on aging found that once a partner's behavior changes (such as physical activity, smoking or losing weight), he/she also changes accordingly.²¹ The pathways from social participation to health outcomes are likely mediated by lifestyle factors.²² Some evidence suggested that a "poor" lifestyle, such as obesity, physical inactivity, unhealthy diet, smoking, and heavy alcohol consumption, is associated with faster cognitive decline in old age.²³ A cross-sectional study suggested that the odds of cognitive impairment was 52% lower among those with healthy versus unhealthy lifestyle.²⁴ A longitudinal study found that adhering to a healthy lifestyle was related to the slow decline of cognitive ability during follow-up.²⁵ In view of the close association between social relationships, lifestyle, and cognitive function, it is reasonable to speculate that healthy lifestyle consequences on social relationships may represent a main potential bridge between social relationships and cognitive function. A few studies have investigated the mediating role of lifestyle factors focusing on only one or two factors. One study suggested that social activities can act on cognitive health through a physical avenue by improving physical activity.¹¹ In addition, cognitive impairment may make it difficult for people to make decisions to maintain a healthy lifestyle, resulting in the decline of social activities and social networks.²⁶ Therefore, lifestyle may also mediate the cognitive function–social relationship pathways. However, compared to the increasing number of studies on the simple one-way social relationship–cognitive function association, very few of them have explored their reciprocal relationship and mechanisms underlying such relationships.

Therefore, the aim of this study was first, to use longitudinal data to examine the reciprocal effects of social relationships and cognitive function. Second, to examine the possible role of healthy lifestyles in the link between social relationships and cognitive function. Based on a review of the literature, the following hypotheses were proposed.

Hypothesis 1. Cognitive function was influenced by social relationships.

Hypothesis 2. Social relationships were influenced by cognitive function.

Hypothesis 3. Healthy lifestyles mediated the association between social relationships and cognitive function.

Method

Participants

The data for this study were drawn from the Chinese Longitudinal Healthy Longevity Study (CLHLS). The CLHLS is a nationwide community-based dynamic longitudinal study aimed at investigating health status and factors influencing longevity in the elderly Chinese population. The survey had been initiated in 1998, and follow-up surveys were conducted in 2000, 2002, 2005, 2008/2009, 2011/2012, 2014, and 2018. Sampling and design of CLHLS are publicly available at <https://sites.duke.edu/centerforaging/>. The CLHLS was approved by the ethics committee of the Peking University (IRB00001052–13074). Written informed consent was obtained from all the participants.

The present analysis included data from the eighth wave in 2014 (at baseline) and the latest wave in 2018. A total of 7,192 older adults were enrolled in 2014. In 2018, 3,441 older adults were re-interviewed, 2,226 died before the re-interview, and 1,525 were lost to follow-up. Finally, the sample included 3,372 participants.

Assessment of cognitive function

The cognitive function was measured using the Chinese version of a modified Mini-Mental State Examination (CMMSE), which has been widely validated in previous studies.^{27,28} The CMMSE consists of 24 items within six dimensions: Five items for Orientation, three for Registration, one for Naming, five for Attention and Calculation, three for Recall, and seven for Language, with a total score ranging from 0 to 30. The lower scores imply poor cognitive functioning. According to previous studies, cognitive impairment was identified if the CMMSE score was less than 24.^{27,28} Changes in cognitive function (Δ cognitive function) as an outcome were calculated as the baseline CMMSE minus the follow-up CMMSE. The CMMSE has been validated in pilot surveys and was comparable to other versions of the MMSE.^{29,30} The Cronbach's alpha for the CMMSE was 0.855 in the cross-sectional analysis of the 2014 study and 0.930 for the follow-up survey.

Assessment of social relationships

Before assessing social relationships, we first evaluated three dimensions: social activities, social networks, and social support. In the current study, social activities included playing cards/mahjong, engaging in organized social activities, and visiting experiences.¹⁴ The scores of social activities ranged from 0 to 3. Four measures of social networks that reflected objective domains were examined, including current marital status, living arrangements, number of children, and siblings who frequently visited the study participant.¹⁴ The scores of social networks ranged from 0 to 4. Social support data were collected with questions on the availability of a person to help with the following situations: (1) when he/she had problems/difficulties; (2) when he/she needed to share some of his/her thoughts; (3) when he/she wanted to talk frequently in daily life; (4) when he/she was sick; (5) when he/she needed financial support, *i.e.*, the amount of monetary support received from his/her children in the past years.¹⁴ The scores of social support ranged from 0 to 6. Details of measurement methods have been described in our previous study.¹⁹

We summed the scores for each dimension to generate a social relationships score ranging from 0 to 13. In addition to taking social relationships as a continuous predictor variable, we also evaluated social relationship levels by tertile (lower, middle, and higher) using lower tertile as the reference group. For further analysis, changes in social relationships (indicated as Δ social relationships) were

calculated as the baseline social relationships minus the follow-up social relationships. The participants were grouped into one of three groups (improved, maintained, or worsened) according to the change in social relationships, and the maintained group was considered the reference group.

Assessment of healthy lifestyle

A healthy lifestyle index was constructed based on five factors, including non-smoking, non-drinking, exercise, healthy diet, and healthy body mass index (BMI). Participants were defined as non-smokers/non-drinkers when they responded to the questions “Are you currently smoking?” and “Do you drink alcohol currently?”, as “No”. Further, participants were asked four questions about types of food, i.e., vegetables, fruits, milk, and tea. We defined a healthy diet if a participant answered “almost every day” or “not every day, but at least once per week” regarding at least two of the four types. BMI was calculated by dividing weight (kg) by height (m^2). If the BMI was greater than 18.5 kg/m^2 and less than 30 kg/m^2 , the BMI was defined as “healthy”.³¹ Participants were given one point if they had at least one healthy lifestyle factor. Further binary scores were added as a healthy lifestyle index, ranging from 0 to 5. The higher the score, the healthier the lifestyle.³¹ Changes in a healthy lifestyle (indicated as Δ the healthy lifestyle index) were calculated as the baseline healthy lifestyle index minus the follow-up healthy lifestyle index.

Covariates

We included potential confounders such as sociodemographic characteristics, lifestyle behaviors, basic activities of daily living (ADL), depressive symptoms, and self-reported chronic diseases, all assessed at baseline in 2014. As described in our previous study,¹⁹ sociodemographic characteristics included age, sex, residence, education level. Lifestyle behaviors included healthy BMI, recent drinking, smoking, healthy diet, exercises. ADL were measured using the ADL scale, encompassing six activities: dressing, eating, toileting, bathing, engaging in indoor activities, and continence. ADL limitations were rated from 0 to 6 and were coded as binary variables (0 = independent, 1 = dependent on one or more ADL limitations). Depressive symptoms were suggested if either of the two questions, i.e., “Have you felt sad, blue, or depressed for two weeks or more in the last 12 months?” and “Have you lost interest in most things like hobbies, work, or similar activities?”, received a response of “yes”. Self-reported chronic diseases include hypertension, diabetes, heart disease, cerebrovascular disease, pulmonary disease, and cancer. The total number of chronic diseases was included in the analysis.

Missing data

We performed multiple imputation by chained equations (MICE) on the full CLHLS 2014–2018 dataset to fill in missing values, consistent with previous studies.^{32,33} MICE is a sequential multivariable regression imputation approach that fills the missing values conditional upon all observed variables.^{32,34,35} Five imputed data sets were produced that included all study variables and were used for this analysis.^{32,35} We calculated the final CMMSE scores, social relationships, and healthy lifestyle index after performing the item-level imputation.

Statistical analyses

Baseline characteristics of participants stratified by cognitive status were described using the mean and standard deviation (SD) of continuous variables and proportions for categorical variables.

Categorical and continuous variables were compared using the χ^2 -test and analysis of variance, respectively.

In the primary analyses, linear regression was initially used to examine the associations between baseline social relationships and the change in the CMMSE score over the follow-up period. Later, binary logistic regression analyses were used to assess whether social relationships at baseline were a predictor of cognition impairment four years later. Three models were developed in this study. Model 1 was adjusted for age, sex, and baseline social relationships as determinants of cognitive function four years later. Model 2 added education, residence, healthy BMI, drinking, smoking, exercise, healthy diet, depressive symptoms, ADL, and the number of chronic diseases, whereas baseline CMMSE score was included in model 3.

In the secondary analyses, a parallel strategy was used to analyze the association between cognitive function and future social relationships. Model 1 was adjusted for age, sex, and baseline cognitive function as determinants of social relationships four years later. Model 2 added education, residence, healthy BMI, drinking, smoking, exercise, healthy diet, depressive symptoms, ADL, and the number of chronic diseases, whereas baseline social relationships score was included in model 3.

Further, the mediation effects of Δ healthy lifestyle on the association of Δ social relationships on Δ cognitive function were tested by model 4, which was analyzed using the PROCESS macro in the SPSS software (IBM Corporation, Armonk, NY, USA).³⁶ Hayes PROCESS estimates model coefficients and produces bias-corrected bootstrap confidence intervals for conditional indirect effects and tests of moderated or conditional mediation (<https://doi.org/10.1177/0002764219859633>). The mediation model could be established if the 95% CI of the indirect effect ($a*b$) did not cross 0. The mediation model was controlled for covariates (age, sex, residence, educational level, depressive symptoms, ADL, and the number of chronic diseases).

In sensitivity analyses, the restricted analysis to people who had no cognition impairment at baseline used the same set of models as above to test the association between baseline social relationships and incident cognitive impairment, and the effects of Δ a healthy lifestyle. The statistically significant level was 0.05 (two-sided). SPSS software version 26.0 (SPSS Inc., Chicago, IL, USA) was used for all analyses.

Results

Characteristics of the participants

Table 1 shows the baseline characteristics of the study participants stratified by cognitive status. At baseline, of the 3,372 participants, 484 (14.4%) had cognitive impairment. Those with cognitive impairment tended to be older females with lower education levels and lower healthy lifestyle index scores, specifically lower levels of healthy BMI and healthy diet, and less physical exercise, smoking, and drinking than those with normal cognition. Furthermore, people with cognitive impairment had higher levels of ADL, depression, and poorer social relationships. During 4-years of follow-up, 514 (17.8%) participants developed incident cognitive impairment (Table S1). There were significant declines in cognitive function, social relationships, and healthy lifestyle index with mean differences of 1.19, 0.36, and 0.07, respectively ($p < 0.001$) (Table S2).

Social relationships as a predictor of future cognitive impairment

Linear regression analysis of the associations between social relationships and changes in cognitive function are summarized in Table 2. In model 1, social relationships at baseline were positively related to changes in cognitive function after adjusting for age and

Table 1

Baseline characteristics of the participants stratified by cognitive status.

Characteristic	Total (n = 3372)	Normal cognition (n = 2888)	Cognitive impairment (n = 484)	P-value
Age (mean ± SD)	81.43±9.03	80.23±8.32	88.57±9.79	<0.001
Sex (%)				<0.001
Male	1606(47.6)	1487(51.5)	119(24.6)	
Female	1766(52.4)	1401(48.5)	365(75.4)	
Education level (%)				<0.001
< 1 year	1717(50.9)	1350(46.7)	367(75.8)	
≥ 1 years	1655(49.1)	1538(53.3)	117(24.2)	
Residence (%)				0.288
Urban	401(11.9)	351(12.2)	50(10.3)	
Town/Rural	2971(88.1)	2537(87.8)	434(89.7)	
Healthy BMI (%)	2772(82.2)	2403(83.2)	369(76.2)	<0.001
Smoking (%)	616(18.3)	557(19.3)	59(12.2)	<0.001
Drinking (%)	601(17.8)	546(18.9)	55(11.4)	<0.001
Exercise (%)	1103(32.7)	988(34.2)	115(23.8)	<0.001
Healthy diet (%)	2212(65.6)	1952(67.6)	260(53.7)	<0.001
Healthy lifestyle index (mean ± SD)	3.44±1.00	3.47±1.01	3.30±0.94	0.001
ADL (%)				<0.001
Independent (with no ADL limitation)	2972(88.1)	2639(91.4)	333(68.8)	
Dependent (with at least one ADL limitation)	400(11.9)	249(8.6)	151(31.2)	
Depressive symptoms (%)	483(14.3)	389(13.5)	94(19.4)	0.001
Number of chronic diseases (mean ± SD)	0.75±1.00	0.76±1.01	0.66±0.98	0.039
Social relationships (mean ± SD)	8.44±1.72	8.59±1.67	7.51±1.71	<0.001
Social activities	0.48±0.73	0.52±0.75	0.23±0.55	<0.001
Social networks	2.68±1.04	2.77±1.03	2.18±0.99	<0.001
Social support	5.28±0.95	5.31±0.92	5.10±1.13	<0.001
CMMSE score (mean ± SD)	26.79±5.06	28.40±1.76	17.23±7.26	<0.001

Notes: SD, standard deviation; BMI, body mass index; ADL, activities of daily living; CMMSE, Chinese Mini-Mental Status Examination.

Dichotomous variables were tested using chi-square tests.

Continuous variables were tested using analysis of variance.

sex ($\beta = 0.043$, $p = 0.018$). In model 2, when demographic variables, lifestyle factors, and other clinical confounders had been taken into account, social relationships were still positively related to changes in cognitive function ($\beta = 0.046$, $p = 0.013$). After adjusting the baseline CMMSE score, social relationships showed a significant negative correlation with cognitive function changes ($\beta = -0.030$, $p = 0.043$).

As shown in Table 3, the binary logistic regression analysis suggested that after covariates were controlled, participants with a 1 SD increase in the social relationships score had an approximately 6% reduced risk of developing cognitive impairment (OR=0.94, 95% CI=0.89–1.00, $p = 0.041$). Similar results were found when modeling the social relationship scores as tertiles. Compared to the lower tertile, the risk of cognitive impairment in the higher tertile social relationships decreased in the fully adjusted model (OR=0.75, 95% CI=0.57

–0.99, $p = 0.045$). The adjusted OR for social relationships increased from 0.67 in model 2 to 0.75 in model 3, indicating that baseline cognitive function explained a substantial proportion of the association between baseline social relationships and future cognitive impairment.

Cognitive function as a predictor of future social relationships

As shown in Table 2, after adjusting for age and sex, the baseline CMMSE scores were positively related to changes in social relationships ($\beta = 0.077$, $p < 0.001$). Further adjustment of associated covariates, the correlation was still significant ($\beta = 0.069$, $p < 0.001$). However, this correlation became insignificant when the baseline social relationships were included in the model ($\beta = -0.005$, $p = 0.735$).

Table 2

Linear regression analysis of the bidirectional associations between social relationships and cognitive function.

Variable	Social relationships as a predictor of changes in cognitive function		Cognitive function as a predictor of changes in social relationships	
	β	P-value	β	P-value
Model 1				
Adjusted for age and sex	0.043	0.018	0.077	<0.001
Model 2				
Additionally adjusted for education, residence, healthy BMI, drinking, smoking, exercise, healthy diet, depressive symptoms, ADL, and number of chronic diseases	0.046	0.013	0.069	<0.001
Model 3				
Additionally adjusted for baseline CMMSE score* or social relationships †	–0.030	0.043	–0.005	0.735

Notes: β , standardized coefficients; BMI, body mass index; ADL, activities of daily living; CMMSE, Chinese Mini-Mental Status Examination.

* Analyses of future cognitive function; † Analyses of future social relationships.

Table 3

Binary logistic regression analysis of baseline social relationships and cognitive impairment at follow-up.

Variable	Model 1		Model 2		Model 3	
	OR (95% CI)	P-value	OR (95% CI)	P-value	OR (95% CI)	P-value
Social relationships						
Per 1-SD increment	0.88(0.84-0.93)	<0.001	0.91(0.86-0.97)	0.002	0.94(0.89-1.00)	0.041
Lower tertile	Ref.		Ref.		Ref.	
Middle tertile	0.83(0.68-1.02)	0.073	0.87(0.71-1.06)	0.171	0.94(0.76-1.16)	0.564
Higher tertile	0.59(0.45-0.77)	<0.001	0.67(0.51-0.89)	0.005	0.75(0.57-0.99)	0.045

Notes: SD, standard deviation; OR, odds ratio; CI, confidence interval.

Model 1: adjusted for age, sex, and baseline social relationships.

Model 2: model 1 plus education, residence, healthy BMI, drinking, smoking, exercise, healthy diet, depressive symptoms, ADL, and the number of chronic diseases.

Model 3: model 2 plus baseline CMMSE score.

Table 4

Binary logistic regression analysis of baseline cognitive function and social relationships at follow-up.

Variable	Worsened group*		Improved group*	
	OR (95% CI)	P-value	OR (95% CI)	P-value
CMMSE score				
Model 1	1.01(0.99-1.03)	0.176	0.98(0.97-1.00)	0.082
Model 2	1.01(0.99-1.03)	0.182	0.99(0.97-1.01)	0.216
Model 3	1.00(0.98-1.02)	0.869	1.01(0.98-1.03)	0.620
Cognitive impairment				
Model 1	0.93(0.72-1.21)	0.579	1.20(0.91-1.59)	0.192
Model 2	0.94(0.72-1.23)	0.645	1.13(0.85-1.50)	0.400
Model 3	1.08(0.82-1.43)	0.593	0.93(0.68-1.25)	0.613

Model 1: adjusted for age, sex, and baseline cognitive function.

Model 2: model 1 plus education, residence, healthy BMI, drinking, smoking, exercise, healthy diet, depressive symptoms, ADL, and the number of chronic diseases.

Model 3: model 2 plus baseline social relationships score.

Notes: OR, odds ratio; CI, confidence interval; CMMSE, Chinese Mini-Mental Status Examination.

*Maintenance group was considered as the reference group.

The multivariate logistic regression analysis suggested that after age and sex were controlled, the baseline cognitive function was not significantly associated with future social relationships, either when analyzed as a continuous variable or classified variable (model 1) improvement group and the worsened group. Further, in the adjustment of other covariates and baseline social relationship scores, the association was still insignificant (model 2 and model 3) (Table 4).

Mediating effects of Δ healthy lifestyles in the association between Δ social relationships and Δ cognitive function

As shown in Table 5, Δ social relationships were associated with Δ healthy lifestyle index ($B = 0.061$, 95% CI = 0.041–0.082). Δ Healthy lifestyle index was associated with Δ cognitive function ($B = 0.413$, 95% CI = 0.226–0.600). The association between Δ social relationships and Δ cognitive function was also significant

($B = 0.219$, 95% CI = 0.105–0.333). Our results also revealed a significant indirect effect of Δ social relationships on Δ cognitive function through Δ healthy lifestyle index ($B = 0.025$, 95% CI = 0.013–0.041). The indirect effect accounted for 10.24% of the total effect, suggesting that Δ healthy lifestyle index only played a partial mediating role in the association between Δ social relationships and Δ cognitive function.

Sensitivity analysis

In Table S3, we present the association between social relationships and incident cognitive impairment. In model 3, which controlled for covariates, participants with a 1 SD increase in the social relationships score had an approximately 7% reduced risk of incident cognitive impairment (OR = 0.93, 95% CI = 0.87–1.00, $p = 0.037$). Compared to the lower tertile, the risk of incident cognitive impairment in the higher tertile social relationships decreased in the fully adjusted model (OR = 0.68, 95% CI = 0.48–0.94, $p = 0.021$). Correspondingly, in Table S4, we present the mediating effects of Δ healthy lifestyle index in the association between Δ social relationships and Δ cognitive function after excluding participants with baseline cognitive impairment. The impacts of Δ social relationships on Δ cognitive function were still partly mediated by Δ healthy lifestyle index.

Discussion

This study examined the bidirectional association between social relationships and cognitive function over a 4-year period. The results demonstrated that positive social relationships might be an important protective factor against cognitive decline, however, the cognitive function has no significant influence on social relationships. In addition, healthy lifestyles played a partial mediating role in the association between social relationships and cognitive function. The findings advanced our understanding of the relationship between the two variables and might contribute to the development of targeted interventions to reduce the risk of cognitive impairment.

Table 5Mediation effects of Δ healthy lifestyle index on the association between Δ social relationships and Δ cognitive function.

Variable	Path	B	SE	LLCI	ULCI
Total effect	Δ Social relationships $\rightarrow$ Δ cognitive function	0.244	0.058	0.130	0.358
Direct effect	Δ Social relationships $\rightarrow$ Δ healthy lifestyle index	0.061	0.011	0.041	0.082
	Δ Healthy lifestyle index $\rightarrow$ Δ cognitive function	0.413	0.095	0.226	0.600
	Δ Social relationships $\rightarrow$ Δ cognitive function	0.219	0.058	0.105	0.333
	Δ Social relationships $\rightarrow$ Δ healthy lifestyle index $\rightarrow$ Δ cognitive function	0.025	0.007	0.013	0.041

Notes: B, unstandardized coefficients; SE, the standard error of indirect effects estimated; LLCI, lower limit confidence interval; ULCI, upper limit confidence interval.

The mediation model was adjusted for age, sex, residence, educational level, depressive symptoms, ADL, and the number of chronic diseases.

The first hypothesis that cognitive function was influenced by social relationships was proven to be desirable. As the previous meta-analysis systematically concluded, poor social relationships represented a lack of social interaction associated with incident dementia or cognitive decline.^{5,37} Even though there are differences in socio-cultural backgrounds, there are similar findings among the sample populations in China, Japan, and South Korea.^{14,38,39} There are clear theories to support the protective effect of positive social relationships on cognitive function. For example, the “use it or lose it” theory and cognitive reserve theory hold that active participation in social activities will stimulate the brain, thus establishing additional neuron resources or cognitive strategies to buffer the decline of cognitive ability.⁷ The social control hypothesis emphasizes that interacting with others in one’s own social network will produce positive and healthy behaviors, which will then affect cognitive function.⁸ In view that cognitive decline is multifactorial in nature, social relationships of different dimensions may have interactive functions. Our study only provided related evidence about the total effects of social relationships on cognitive functioning. It is necessary to actively explore how different dimensions of social relationships affect cognitive function in the future.

Social relationships often change and decrease in later life, especially the reduction of social activities because older adults have limited mobility or feel self-conscious about worsening memory or increasing forgetfulness.⁴⁰ In our study, the scores of social networks, social activities, and social support of people with cognitive impairment were lower than those with normal cognition. Therefore, reverse causality cannot be excluded. This is also a problem that the existing meta-analysis cannot solve but has been advocating.^{5,37} In a population-based study that used data from 13,984 people in Japan, Saito *et al.* reported that respondents engaging in five social relationship domains were less likely to develop dementia by almost 50%.³⁸ However, the study emphasized that they cannot deny the possibility of reverse causation. In our study, baseline cognitive function did not independently predict social relationships four years later, even if we excluded the baseline cognitive impairment, it was still a similar result. Another study evaluated the bidirectional causal relationship, and a similar conclusion was reached.³⁹ The absence of this reverse causality may be explained by the following aspects. On the one hand, with the decline of cognitive function, the peripheral connection may be shed while the closest social connection may be maintained.⁴¹ On the other hand, cognitive decline will lead to the loss of social contact but will also lead to an increase in instrumental support. Older people who experience cognitive decline begin to get more nursing-related support. This kind of support may be mainly instrumental, including physical and financial assistance, but may also include emotional support. Moreover, individuals accessing varied services may build up social networks that can contribute to building cognitive reserve.⁴² In short, the current evidence does not support the second hypothesis that the decline of cognitive function leads to poor social relationships, and in the future, longitudinal randomized controlled trials are needed to verify the causal relationship between the two.

Previous research suggested that social relationships were likely intertwined with other health-related lifestyle factors that may have a synergistic effect on cognitive decline.⁴³ Our analyses indicated that changes in social relationships and healthy lifestyles were associated with cognitive function. Furthermore, a healthy lifestyle may mitigate the cognitive impairment caused by poor social relationships. A cohort study based on the European population also suggested that people who were socially active engaged in more physical activity over time, which was related to better subsequent cognitive function.¹¹ A possible explanation for the mediational pathway is that social resources may yield multiple sources of information that contribute to an effective positive healthy afternoon.²⁵ Furthermore,

positive social relationships and a healthy lifestyle are conducive to a positive psychological state, such as a sense of belonging and meaning. This may modulate the neuroendocrine response to stress affecting the brain.⁴³ In fact, social relationships and lifestyle seem to have common pathways rather than specific mechanisms, which might converge within three major aetiological hypotheses for dementia: cognitive reserve, vascular, and stress hypothesis.⁴³ Further research on the relationship between social relationships and the maintenance of cognitive function may benefit from attention to different aspects of lifestyles.

Interpretation of our results should be considered within the context of a number of limitations of the study. First, cognitive function was only assessed by CMMSE, without clinical assessment or other cognitive tests. CMMSE is a simple measurement of the overall cognitive function, which may not be sensitive enough to detect changes in cognitive function. Second, social relationships and lifestyle factors were evaluated by single-question and self-reported items on a questionnaire. As a result, they may suffer from justification bias and misclassifications. The lack of a universal and available scale of social relationships in the CLHS may limit our findings compared with other studies. Third, the average follow-up time of this study was four years, which was relatively short. The longer the follow-up time, the stronger the association between cognitive function and future social relationships may be observed. Fourth, although we used some representative large-scale national survey data, the outdated data made our findings more suitable for the period before the outbreak of COVID-19. After the virus outbreak, the social relationships of the elderly in China are very different from those in the past, which will lead to limited generalization of our findings.

In summary, this study explored the bidirectional associations between social relationships and cognitive function in China using nationally representative data. Our results showed that positive social relationships might be an important protective factor against cognitive decline, but cognitive function had no significant influence on social relationships. In addition, lifestyle factors influenced the mechanisms linking social relationships and cognitive function. Our results suggested that social relationships-based cognitive function policies would benefit from taking lifestyle factors into account.

Data Availability Statement

The original contributions presented in this study are included in the article/supplementary material, further inquiries can be directed to the corresponding authors.

Ethics Statement

The studies involving human participants were reviewed and approved by the ethics committee of the Peking University (IRB00001052–13074). The participants provided their written informed consent to participate in this study.

Author Contributions

WM and BW conceived the concept and design of the study; WM contributed to data cleaning and analysis. WM, YY and RZ contributed to the writing assistance and proof reading the article. All authors approved the final version of the manuscript.

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Declaration of Competing Interest

The authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

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Supplementary materials

Supplementary material associated with this article can be found in the online version at doi:[10.1016/j.gerinurse.2023.06.016](https://doi.org/10.1016/j.gerinurse.2023.06.016).

References

- Chowdhary N, Barbui C, Anstey KJ, et al. Reducing the risk of cognitive decline and dementia: who recommendations. *Front Neurol*. 2021;12: 765584. <https://doi.org/10.3389/fneur.2021.765584>.
- World Health Organization. *Global Status Report on the Public Health Response to Dementia 2017–2025*. Geneva: World Health Organization; 2021.
- Jia L, Quan M, Fu Y, et al. Dementia in China: epidemiology, clinical management, and research advances. *Lancet Neurol*. 2020;19(1):81–92. [https://doi.org/10.1016/S1474-4422\(19\)30290-X](https://doi.org/10.1016/S1474-4422(19)30290-X).
- Jia L, Du Y, Chu L, et al. Prevalence, risk factors, and management of dementia and mild cognitive impairment in adults aged 60 years or older in China: a cross-sectional study. *Lancet Public Health*. 2020;5(12):e661–e671. [https://doi.org/10.1016/S2468-2667\(20\)30185-7](https://doi.org/10.1016/S2468-2667(20)30185-7).
- Kuiper JS, Zuidersma M, Zuidema SU, et al. Social relationships and cognitive decline: a systematic review and meta-analysis of longitudinal cohort studies. *Int J Epidemiol*. 2016;45(4):1169–1206. <https://doi.org/10.1093/ije/dyw089>.
- Kelly ME, Duff H, Kelly S, et al. The impact of social activities, social networks, social support and social relationships on the cognitive functioning of healthy older adults: a systematic review. *Syst Rev*. 2017;6(1):259. <https://doi.org/10.1186/s13643-017-0632-2>.
- Fisher GG, Chaffee DS, Tetrick LE, Davalos DB, Potter GG. Cognitive functioning, aging, and work: a review and recommendations for research and practice. *J Occup Health Psychol*. 2017;22(3):314–336. <https://doi.org/10.1037/ocp0000086>.
- Umberson D. Family status and health behaviors: social control as a dimension of social integration. *J Health Soc Behav*. 1987;28(3):306–319.
- Cohen S, Wills TA. Stress, social support, and the buffering hypothesis. *Psychol Bull*. 1985;98(2):310–357.
- Kuiper JS, Zuidersma M, Oude VR, et al. Social relationships and risk of dementia: a systematic review and meta-analysis of longitudinal cohort studies. *Ageing Res Rev*. 2015;22:39–57. <https://doi.org/10.1016/j.arr.2015.04.006>.
- Cohn-Schwartz E. Pathways from social activities to cognitive functioning: the role of physical activity and mental health. *Innov Aging*. 2020;4(3):a15. <https://doi.org/10.1093/geronl/igaa015>.
- Yin S, Yang Q, Xiong J, Li T, Zhu X. Social support and the incidence of cognitive impairment among older adults in China: findings from the Chinese longitudinal healthy longevity survey study. *Front Psychiatry*. 2020;11:254. <https://doi.org/10.3389/fpsy.2020.00254>.
- Zhou Z, Wang P, Fang Y. Social engagement and its change are associated with dementia risk among Chinese older adults: a longitudinal study. *Sci Rep*. 2018;8(1):1551. <https://doi.org/10.1038/s41598-017-17879-w>.
- Fan Z, Lv X, Tu L, Zhang M, Yu X, Wang H. Reduced social activities and networks, but not social support, are associated with cognitive decline among older Chinese adults: a prospective study. *Soc Sci Med*. 2021;289: 114423.
- Lindau ST, Laumann EO, Levinson W, Waite LJ. Synthesis of scientific disciplines in pursuit of health: the interactive biopsychosocial model. *Perspect Biol Med*. 2003;46(3 Suppl):S74–S86. <https://doi.org/10.1353/pbm.2003.0055>.
- Xu M, Thomas PA, Umberson D. Marital quality and cognitive limitations in late life. *J Gerontol B*. 2016;71(1):165–176. <https://doi.org/10.1093/geronb/gbv014>.
- Wong JS, Hsieh N. Functional status, cognition, and social relationships in dyadic perspective. *J Gerontol B*. 2019;74(4):703–714. <https://doi.org/10.1093/geronb/gbx024>.
- Perlmutter MS, Bhorade A, Gordon M, Hollingsworth HH, Baum MC. Cognitive, visual, auditory, and emotional factors that affect participation in older adults. *Am J Occup Ther*. 2010;64(4):570–579. <https://doi.org/10.5014/ajot.2010.09089>.
- Ma W, Wu B, Gao X, Zhong R. Association between frailty and cognitive function in older Chinese people: a moderated mediation of social relationships and depressive symptoms. *J Affect Disord*. 2022;316:223–232. <https://doi.org/10.1016/j.jad.2022.08.032>.
- Asamane EA, Greig CA, Thompson JL. Social networks and their influences on nutrient intake, nutritional status and physical function in community-dwelling ethnically diverse older adults: a mixed-methods longitudinal study. *Bmc Public Health*. 2020;20(1):1011. <https://doi.org/10.1186/s12889-020-09153-y>.
- Jackson SE, Steptoe A, Wardle J. The influence of partner's behavior on health behavior change: the English longitudinal study of ageing. *JAMA Intern Med*. 2015;175(3):385–392. <https://doi.org/10.1001/jamainternmed.2014.7554>.
- Ziegelmann JP, Knoll N. Future directions in the study of health behavior among older adults. *Gerontology*. 2015;61(5):469–476. <https://doi.org/10.1159/000369857>.
- Wajman JR, Mansur LL, Yassuda MS. Lifestyle patterns as a modifiable risk factor for late-life cognitive decline: a narrative review regarding dementia prevention. *Curr Aging Sci*. 2018;11(2):90–99. <https://doi.org/10.2174/1874609811666181003160225>.
- Jin X, He W, Zhang Y, et al. Association of APOE epsilon4 genotype and lifestyle with cognitive function among Chinese adults aged 80 years and older: a cross-sectional study. *Plos Med*. 2021;18(6): e1003597. <https://doi.org/10.1371/journal.pmed.1003597>.
- Dhana K, Aggarwal NT, Rajan KB, Barnes LL, Evans DA, Morris MC. Impact of the apolipoprotein E epsilon4 allele on the relationship between healthy lifestyle and cognitive decline: a population-based study. *Am J Epidemiol*. 2021;190(7):1225–1233. <https://doi.org/10.1093/aje/kwab033>.
- Mann T, de Ridder D, Fujita K. Self-regulation of health behavior: social psychological approaches to goal setting and goal striving. *Health Psychol*. 2013;32(5):487–498. <https://doi.org/10.1037/a0028533>.
- Zhang Q, Wu Y, Han T, Liu E. Changes in cognitive function and risk factors for cognitive impairment of the elderly in China: 2005–2014. *Int J Environ Res Public Health*. 2019;16(16). <https://doi.org/10.3390/ijerph16162847>.
- Zeng Y, Feng Q, Hesketh T, Christensen K, Vaupel JW. Survival, disabilities in activities of daily living, and physical and cognitive functioning among the oldest-old in China: a cohort study. *Lancet*. 2017;389(10079):1619–1629. [https://doi.org/10.1016/S0140-6736\(17\)30548-2](https://doi.org/10.1016/S0140-6736(17)30548-2).
- Zeng Y. Towards deeper research and better policy for healthy aging –using the unique data of Chinese longitudinal healthy longevity survey. *China Economic J*. 2012;5(2–3):131–149. <https://doi.org/10.1080/17538963.2013.764677>.
- Zhang Z, Gu D, Hayward MD. Early life influences on cognitive impairment among oldest old Chinese. *J Gerontol B*. 2008;63(1):S25–S33. <https://doi.org/10.1093/geronb/63.1.s25>.
- Jin S, Li C, Cao X, Chen C, Ye Z, Liu Z. Association of lifestyle with mortality and the mediating role of aging among older adults in China. *Arch Gerontol Geriatr*. 2022;98: 104559. <https://doi.org/10.1016/j.archger.2021.104559>.
- Gao M, Kuang W, Qiu P, Wang H, Lv X, Yang M. The time trends of cognitive impairment incidence among older Chinese people in the community: based on the CLHLS cohorts from 1998 to 2014. *Age Ageing*. 2017;46(5):787–793. <https://doi.org/10.1093/ageing/afx038>.
- Oginni OA, Robinson EJ, Jones A, Rahman Q, Rimes KA. Mediators of increased self-harm and suicidal ideation in sexual minority youth: a longitudinal study. *Psychol Med*. 2019;49(15):2524–2532. <https://doi.org/10.1017/S003329171800346X>.
- Azur MJ, Stuart EA, Frangakis C, Leaf PJ. Multiple imputation by chained equations: what is it and how does it work? *Int J Methods Psychiatr Res*. 2011;20(1):40–49. <https://doi.org/10.1002/mpr.329>.
- Godin J, Keefe J, Andrew MK. Handling missing mini-mental state examination (MMSE) values: results from a cross-sectional long-term-care study. *J Epidemiol*. 2017;27(4):163–171. <https://doi.org/10.1016/j.je.2016.05.001>.
- Hayes F. In: *Process: A Versatile Computational Tool for Observed Variable Mediation, Moderation, and Conditional Process Modeling*. 2012:1–17. [white paper]; <http://www.afhayes.com/public/process2012.pdf>. 20.
- Piolatto M, Bianchi F, Rota M, Marengoni A, Akbaritabar A, Squazzoni F. The effect of social relationships on cognitive decline in older adults: an updated systematic review and meta-analysis of longitudinal cohort studies. *BMC Public Health*. 2022;22(1):278. <https://doi.org/10.1186/s12889-022-12567-5>.
- Saito T, Murata C, Saito M, Takeda T, Kondo K. Influence of social relationship domains and their combinations on incident dementia: a prospective cohort study. *J Epidemiol Commun Health*. 2018;72(1):7–12. <https://doi.org/10.1136/jech-2017-209811>.
- Park S, Downer B, Nam S, Bae S, Hong I. Association between social participation and cognitive function in community-dwelling adults. *Otjir*. 2022;42(4):344–352. <https://doi.org/10.1177/15394492221113506>.
- Rotenberg S, Sternberg S, Maier A. Where did I put my glasses? The lived experience of older adults seeking medical help for perceived memory problems. *Disabil Rehabil*. 2020;42(25):3606–3613. <https://doi.org/10.1080/09638288.2019.1602849>.
- Clay OJ, Roth DL, Wadley VG, Haley WE. Changes in social support and their impact on psychosocial outcome over a 5-year period for African American and white dementia caregivers. *Int J Geriatr Psychiatry*. 2008;23(8):857–862. <https://doi.org/10.1002/gps.1996>.
- Siette J, Georgiou A, Brayne C, Westbrook JL. Social networks and cognitive function in older adults receiving home- and community-based aged care. *Arch Gerontol Geriatr*. 2020;89: 104083.
- Fratiglioni L, Paillard-Borg S, Winblad B. An active and socially integrated lifestyle in late life might protect against dementia. *Lancet Neurol*. 2004;3(6):343–353. [https://doi.org/10.1016/S1474-4422\(04\)00767-7](https://doi.org/10.1016/S1474-4422(04)00767-7).